

Modeling of Vaccine Tender Problem

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1 Two-Stage Stochastic Program Mathematical Formulation - Quantity Discount

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1.1 Producer capacity-based quantity discount

The capacity discount model includes quantity based pricing discounts based on discount segments, m , and is applied to instances of $X_{vpt\omega}$ in the model, now updated as $X_{vptm\omega}$.

1.1.1 Model Notation

Sets:

- $\mathcal{V}$: set of vaccines (from the WHO pre-qualified vaccine list),
- $\mathcal{V}_a$: subset of vaccines containing antigen a ,
- $\mathcal{A}$: set of antigens,
- $\mathcal{A}_v$: subset of antigens in vaccine v ,
- $\mathcal{A}_p$: subset of antigens contained in producer p 's portfolio,
- $\mathcal{P}$: set of producers,
- $\mathcal{P}_v$: subset of producers of vaccine v ,
- $\mathcal{T}$: set of time periods in the planning horizon,
- $\mathcal{T}'_t$: subset of periods when a tender ends (since we allow tenders of at most Δ years, $\mathcal{T}'_t = \{t, t+1, \dots, \min\{t+\Delta, |\mathcal{T}|\}\}$).
- Ω : set of scenarios for uncertain parameters

Parameters:

- $\tilde{d}_{at\omega}$: stochastic demand at time t for antigen a in scenario ω ,
- $\tilde{s}_{pt\omega}$: stochastic production capacity of producer p at time t in scenario ω ,
- $\bar{s}_p$: base production capacity of producer p ,
- r_{vpt} : the reservation price of vaccine v produced by p in period t ,
- $\bar{r}_{vt}$: average price of vaccine v across all producers in period t ,
- l_{vp} : the annualized return on investment required by producer p for vaccine v ,

- g_t : set-up cost for initiating a tender in period t ,
- f_{vpt} : the production set-up cost of producer p for vaccine v in period t ,
- h_v : the annual holding cost for vaccine v as a proportion of its price,
- β : penalty per unit of shortage of the committed amount, i.e., the missed dose penalty,
- Γ_p : capacity extension cost for producer p ,
- κ : allowable capacity extension rate for producer p ,
- n : maximum value that capacity extension variable, L_{pt} , can get.
- δ_t : a discount factor to capture the time value of money
- Δ : maximum allowable tender length.
- $\hat{I}_{v0}$: initial inventory level for vaccine v ,
- $\hat{S}_{a0}$: initial unvaccinated children condition of antigen a ,
- $\hat{F}_{a0\tau}$: ongoing tenders at the beginning of the planning horizon for antigen a by the period τ ,
- $\hat{Q}_{vp0\tau}$: commitment amount for the ongoing tender of vaccine v by producer p by the period τ .

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Updated Parameters:

- $\gamma_{ptm\omega}$: discount level per dose achieved for producer p in period t discount segment m , scenario ω ,
- $q_{ptm\omega}^L$: the lower bound of the vaccine delivery quantity for producer p in period t and discount segment m , scenario ω ,
- $q_{ptm\omega}^U$: the upper bound of the vaccine delivery quantity for producer p in period t and discount segment m , scenario ω ,
- θ_m^L : the lower bound of the capacity usage rate for discount segment m ,
- θ_m^U : the upper bound of the capacity usage rate for discount segment m

$$q_{ptm\omega}^L = \bar{s}_{pt}\theta_m^L$$

$$q_{ptm\omega}^U = \bar{s}_{pt}\theta_m^U$$

Decision Variables:

- $F_{at\tau}$: a binary variable indicating whether a tender covers the demand for antigen a for periods t through τ ,
- Y_{pt} : a binary variable which specifies whether or not manufacturer p produces in period t ,
- $W_{pt\tau}$: a binary variable indicating if producer p has made commitments for periods t through τ .
- L_{pt} : an integer variable representing capacity extension for producer p in period t .
- $Q_{vpt\tau}$: procurement commitment for vaccine v by producer p for periods t through τ ,
- $X_{vpt\omega}$: doses of vaccine v delivered by producer p in period t for scenario ω ,

- $I_{vt\omega}$: stock level for vaccine v at the beginning of period t (including period 0) for scenario ω ,
- $V_{vt\omega}$: number of doses administered with vaccine v in period t for scenario ω ,
- $S_{at\omega}$: number of missed doses for antigen a in period t (including period 0) for scenario ω .

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Updated Variables:

- $X_{vptm\omega}$: doses of vaccine v delivered by producer p in period t at discount quantity m for scenario ω ,

1.1.2 Model Formulation

The first stage of the two-stage stochastic program model formulation is given as (1):

$$\min \sum_{t \in \mathcal{T}} \delta_t \left(\sum_{a \in \mathcal{A}} \sum_{\tau \in \mathcal{T}'_t} g_t F_{at\tau} + \sum_{p \in \mathcal{P}} \Gamma_p L_{pt} \right) + \mathbb{E}_\Omega[\mathcal{Q}(\bar{\Xi}, \omega)] \quad (1)$$

$$(\tau - t + 1)F_{at\tau} \leq \sum_{p \in \mathcal{P}_a} \sum_{l=t}^{\tau} Y_{pl}, \quad \forall a \in \mathcal{A}; t \in \mathcal{T}; \tau \in \mathcal{T}'_t. \quad (2)$$

$$F_{at\tau} + F_{at\tau'} \leq 1, \quad \forall a \in \mathcal{A}; t \in \mathcal{T}; \tau, \tau' \in \mathcal{T}'_t; \tau \neq \tau'. \quad (3)$$

$$\sum_{t=\max\{1, l-\Delta\}}^l \sum_{\tau \in \mathcal{T}'_t} F_{at\tau} \geq 1, \quad \forall a \in \mathcal{A}; l \in \mathcal{T}. \quad (4)$$

$$\sum_{a \in \mathcal{A}_p} F_{at\tau} \geq W_{pt\tau}, \quad \forall p \in \mathcal{P}; t \in \mathcal{T}; \tau \in \mathcal{T}'_t. \quad (5)$$

$$\sum_{a \in \mathcal{A}_p} F_{at\tau} \leq |\mathcal{A}_p| W_{pt\tau}, \quad \forall p \in \mathcal{P}; t \in \mathcal{T}; \tau \in \mathcal{T}'_t. \quad (6)$$

$$\sum_{v \in \mathcal{V}_p} Q_{vpt\tau} \leq W_{pt\tau} \left(\sum_{l=t}^{\tau} (\bar{s}_p + \sum_{k=1}^l \kappa \bar{s}_p L_{pk}) \right), \quad \forall p \in \mathcal{P}; t \in \mathcal{T}; \tau \in \mathcal{T}'_t. \quad (7)$$

$$F_{at\tau}, Y_{pt}, W_{pt\tau} \in \{0, 1\}, L_{pt} \in \{0, \dots, n\}, Q_{vpt\tau} \geq 0 \quad (8)$$

NU: The second stage of the two-stage stochastic program model (9) is modified to consider quantity discounts. The inclusion of the parameter $(1 - \gamma_m)$ in the sub problem objective function, allows the model to include a discount pricing scheme, and consider whole unit quantity discounts.

$$\mathcal{Q}(\bar{\Xi}, \omega) = \min \sum_{t \in \mathcal{T}} \delta_t \left(\sum_{v \in \mathcal{V}} \sum_{p \in \mathcal{P}_v} (1 - \gamma_{m\omega}) r_{vpt} X_{vptm\omega} + \sum_{a \in \mathcal{A}} \beta S_{at\omega} + \sum_{v \in \mathcal{V}} h_v \bar{r}_{vt} I_{vt\omega} \right) \quad (9)$$

$$Q_{vpt\tau} \geq W_{pt\tau} \sum_{l=t}^{\tau} X_{vplm\omega}, \quad \forall v \in \mathcal{V}, p \in \mathcal{P}_v, t \in \mathcal{T}, \tau \in \mathcal{T}'_t, \quad (10)$$

$$\sum_{v \in \mathcal{V}_p} X_{vptm\omega} \leq Y_{pt} (\tilde{s}_{pt\omega} + \sum_{l=1}^t \kappa \bar{s}_p L_{pl}), \quad \forall p \in \mathcal{P}; t \in \mathcal{T}, \quad (11)$$

$$I_{vt-1, \omega} + \sum_{p \in \mathcal{P}_v} X_{vptm\omega} = V_{vt\omega} + I_{vt\omega}, \quad \forall v \in \mathcal{V}; t \in \mathcal{T}. \quad (12)$$

$$\tilde{d}_{at\omega} - \sum_{v \in \mathcal{V}_a} V_{vt\omega} + S_{at-1,\omega} \leq S_{at\omega}, \quad \forall a \in \mathcal{A}; t \in \mathcal{T}. \quad (13)$$

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$$\sum_{v \in \mathcal{V}_p} (1 - \gamma_{m\omega}) r_{vpt} X_{vptm\omega} \geq \sum_{v \in \mathcal{V}_p} (1 + l_{vp}) f_{vpt} Y_{pt}, \quad \forall p \in \mathcal{P}; t \in \mathcal{T}. \quad (14)$$

$$I_{v0\omega} = \hat{I}_{v0}, \quad \forall v \in \mathcal{V}. \quad (15)$$

$$S_{a0\omega} = \hat{S}_{a0}, \quad \forall a \in \mathcal{A}. \quad (16)$$

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$$X_{vptm\omega}, K_{vpt\tau\omega}, I_{vt\omega}, S_{at\omega}, V_{vt\omega} \geq 0 \quad (17)$$

Constraint (10) includes a non-convex bi-linear term, where a binary variable ($W_{pt\tau}$) and summation of continuous variables ($\sum_{l=t}^{\tau} X_{vplm\omega}$) are multiplied. We linearize (10) by using the well-known McCormick envelopes [1]. Note that $\langle xy \rangle^M$ in equation (18) denotes the variable representing the bilinear term (xy). When $\langle xy \rangle^M$ is incorporated into an equation, the corresponding convex envelopes are also integrated into the model. Note that $\hat{X}_{vptm\tau\omega}$ is a continuous variable that represents $\sum_{l=t}^{\tau} X_{vplm\omega}$, which is used in McCormick relaxation for simplicity. In the reformulation, constraint (10) is replaced with the following constraint sets (19) - (20). This is a tight relaxation since $W_{pt\tau}$ is a binary variable.

$$\langle xy \rangle^M \equiv \begin{cases} \tilde{xy} \leq x^l y + xy^l - x^l y^l \\ \tilde{xy} \leq x^u y + xy^u - x^u y^u \\ \tilde{xy} \geq x^l y + xy^u - x^l y^u \\ \tilde{xy} \geq x^u y + xy^l - x^u y^u \end{cases} \quad (18)$$

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$$\hat{X}_{vptm\tau\omega} = \sum_{l=t}^{\tau} X_{vplm\omega}, \quad \forall v \in \mathcal{V}; p \in \mathcal{P}_v; t \in \mathcal{T}; \tau \in \mathcal{T}'_t. \quad (19)$$

$$Q_{vpt\tau} \geq \langle W_{pt\tau} \hat{X}_{vptm\tau\omega} \rangle^M, \quad \forall v \in \mathcal{V}; p \in \mathcal{P}_v; t \in \mathcal{T}; \tau \in \mathcal{T}'_t. \quad (20)$$

Constraint (7) also includes non-convex bi-linear terms, where binary variables ($W_{pt\tau}$) and (L_{pt}) are multiplied. We linearize (7) by using McCormick envelopes. However, first, we convert nested summations in equation (7) to individual summations with a new representation in equation (21). Note that Equations (7) and (21) are mathematically equal to each other.

$$\sum_{v \in \mathcal{V}_p} Q_{vpt\tau} \leq W_{pt\tau} \left(\sum_{l=t}^{\tau} \bar{s}_p + \sum_{l=t+1}^{\tau} (\tau - l + 1) \kappa \bar{s}_p L_{pl} + \sum_{l=1}^t (\tau - t + 1) \kappa \bar{s}_p L_{pl} \right), \quad \forall p \in \mathcal{P}; t \in \mathcal{T}; \tau \in \mathcal{T}'_t. \quad (21)$$

For simplicity, we define two new variables ($\hat{L}_{pt\tau}$ and $\check{L}_{pt\tau}$) representing the following summations ($\sum_{l=t+1}^{\tau} (\tau - l + 1) \kappa \bar{s}_p L_{pl}$) and ($\sum_{l=1}^t (\tau - t + 1) \kappa \bar{s}_p L_{pl}$), respectively. Then, we can rewrite equation (21) as follows:

$$\sum_{v \in \mathcal{V}_p} Q_{vpt\tau} \leq W_{pt\tau} \left(\sum_{l=t}^{\tau} \bar{s}_p + \hat{L}_{pt\tau} + \check{L}_{pt\tau} \right), \quad \forall p \in \mathcal{P}; t \in \mathcal{T}; \tau \in \mathcal{T}'_t. \quad (22)$$

There are two bi-linear terms in equation (22) where a binary variable, $W_{pt\tau}$, is multiplied by the newly defined variables, $\hat{L}_{pt\tau}$ and $\check{L}_{pt\tau}$. We replace constraint (22) with the following set of constraints (23) - (25):

$$\hat{L}_{pt\tau} = \sum_{l=t+1}^{\tau} (\tau - l + 1) \kappa \bar{s}_p L_{pl}, \quad \forall p \in \mathcal{P}; t \in \mathcal{T}; \tau \in \mathcal{T}'_t. \quad (23)$$

$$\check{L}_{pt\tau} = \sum_{l=1}^t (\tau - t + 1) \kappa \bar{s}_p L_{pl}, \quad \forall p \in \mathcal{P}; t \in \mathcal{T}; \tau \in \mathcal{T}'_t. \quad (24)$$

$$\sum_{v \in \mathcal{V}_p} Q_{vpt\tau} \leq W_{pt\tau} \left(\sum_{l=t}^{\tau} \bar{s}_p \right) + \langle W_{pt\tau} \hat{L}_{pt\tau} \rangle^M + \langle W_{pt\tau} \check{L}_{pt\tau} \rangle^M, \quad \forall p \in \mathcal{P}; t \in \mathcal{T}; \tau \in \mathcal{T}'_t. \quad (25)$$

Similarly, we linearize the non-convex bi-linear term in equation (11) using McCormick envelopes. Let $\ddot{L}_{pt}$ represents the summation $(\sum_{l=1}^t \kappa \bar{s}_p L_{pl})$. We replace constraint (11) with the following constraints:

$$\ddot{L}_{pt} = \sum_{l=1}^t \kappa \bar{s}_p L_{pl}, \quad \forall p \in \mathcal{P}; t \in \mathcal{T}. \quad (26)$$

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$$\sum_{v \in \mathcal{V}_p} X_{vptm\omega} \leq Y_{pt} \ddot{L}_{pt\omega} + \langle Y_{pt} \ddot{L}_{pt} \rangle^M, \quad \forall p \in \mathcal{P}; t \in \mathcal{T}, \quad (27)$$

The following model summarizes the final version of the DE model:

$$DE \text{ first-stage} : \min \sum_{t \in \mathcal{T}} \delta_t \left(\sum_{a \in \mathcal{A}} \sum_{\tau \in \mathcal{T}'_t} g_t F_{at\tau} + \sum_{p \in \mathcal{P}} \Gamma_p L_{pt} \right) + \sum_{\omega \in \Omega} p_{\omega} \mathcal{Q}(\bar{\Xi}, \omega)$$

constraints (2) - (6), (8), (23) - (26)

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$$DE \text{ second-stage} : \mathcal{Q}(\bar{\Xi}, \omega) = \min \sum_{t \in \mathcal{T}} \delta_t \left(\sum_{v \in \mathcal{V}} \sum_{p \in \mathcal{P}_v} (1 - \gamma_{m\omega}) r_{vpt} X_{vptm\omega} + \sum_{a \in \mathcal{A}} \beta S_{at\omega} + \sum_{v \in \mathcal{V}} h_v \bar{r}_{vt} I_{vt\omega} \right)$$

constraints (12) - (17), (19) - (20), (27)

References

- [1] G. P. McCormick, “Computability of global solutions to factorable nonconvex programs: Part i—convex underestimating problems,” *Mathematical programming*, vol. 10, no. 1, pp. 147–175, 1976.